Institutional Performance Audit — Spot Neutral Grid Program

Source: `Grid_Bot_Tracker_Excel_1.xlsx` (Setup · Bot Log · Weekly Review) **Venue:** Pionex (workbook subtitle says "Bybit" — see Data Integrity) **Audit window:** 2026-09-20 → 2026-10-21 (32 calendar days) · **Starting capital:** 1,000 USDT **Prepared:** 2026-10-22

Scope note — read first. The journal contains *realized, closed-bot results only* (entry/exit dates, grid range, grid count, investment, final value). It contains **no price series, no fill-level data, no fee line items, no open-bot marks and no intraday timestamps.** Every metric below is labelled **[Measured]** (computed from the journal), **[Derived]** (model-based, assumptions stated) or **[Not computable]** (data absent). Nothing marked Measured has been estimated, and I have not invented ATR, slippage, or Buy-and-Hold figures.

Start here: the report in plain English

What was audited? Your grid-bot journal: 34 closed bots on Pionex between Sep 20 and Oct 21, 2026, starting with 1,000 USDT. A grid bot places buy and sell orders in a price band and earns small profits each time price bounces up and down.

- **The good:** +322.94 USDT (+32.3%) in 32 days. 33 of 34 bots made money. Weekly returns were steady (4.7%–7.7%).
- **The catch:** the journal only shows *finished* bots. It cannot show losses sitting in bots that were still open or price drops you avoided by closing early. So "0% drawdown" and "Sharpe 12.6" look perfect but are **not proof of safety**.
- **The biggest question:** on Oct 20 about 3,874 USDT was logged as invested while the account was worth about 1,300. Please check whether that was leverage, bots run back-to-back the same day, or typing errors.
- **The fix list:** remove the "EXAMPLE ROW" from totals (profit then = 315.39), log fees and prices, and spread money away from low-paying pairs (TRX, INJ).

Rating: 6.5 / 10 — strong results, but too short and too incomplete to call proven.

Glossary of every parameter used

Term	Meaning
Grid / grid count	Number of price levels where the bot places orders. More grids = smaller steps, more trades, more fees.
Range (lower / upper)	Price band the bot works inside. If price leaves it, the bot stops earning.
Step (spacing)	Percentage gap between grid levels. Must be wider than round-trip fees to profit.
Neutral mode	Bot starts about half in coin, half in USDT and profits from bounce either way.
ROI %	Profit ÷ investment for one bot.

Win rate / Profit Factor	Share of profitable bots / total winnings ÷ total losses (73.6 means $73.60 won per $1 lost).
Drawdown (MDD)	Largest fall from a previous equity peak. Here only measured on realized profit, so it reads 0%.
Sharpe / Sortino / Calmar	Return per unit of risk (all swings / downside swings only / worst drawdown). Unreliable on 32 days.
CAGR / annualized ROI	Return projected to a full year. Not meaningful from one month.
Inventory skew	When price falls a neutral grid ends up holding mostly coin (long-heavy) — the source of its loss risk.
Out-of-range risk	Price moving below the lower bound (bot stuck holding coin) or above the upper bound (bot fully sold, earning nothing).
Impermanent / divergence loss	How much worse the bot does than simply holding the coin when price trends.
Fee drag / slippage	Profit lost to trading fees / to fills at worse prices than expected.
Capital efficiency	How much of your money is working versus idle.
ATR	Average True Range: typical price swing per period; used to size grid steps.
Delta hedge	Opening an offsetting short position so price moves hurt less.
[Measured] / [Derived] / [Not computable]	Taken straight from your data / modelled with stated assumptions / impossible without missing data.

Section I — Executive Summary

Overall Rating: **6.5 / 10 — *Promising edge, unproven robustness* (provisional)**

Component	Weight	Score	Rationale
Realized return	25%	9.0	+32.3% in 32 days; 33 of 34 bots profitable
Risk-adjusted quality	25%	5.0	Ratios are mathematically excellent but **statistically meaningless** (n = 32 days, realized-only, zero MTM drawdown observable)
Capital efficiency	15%	7.0	~80% average deployment; but capital is concentrated in the *lowest-yielding* pairs (TRX, INJ)
Grid robustness / boundary risk	20%	5.0	No out-of-range event recorded; single-name concentration (NEAR = 43% of profit)
Data integrity	15%	5.5	Example row in totals, orphan rows, concurrency inconsistency, no fee data
Weighted total		**6.5**	Rating is *capped by evidence quality*, not by performance

High-Level Performance Snapshot

Metric	Value	Basis
Total realized profit (in-window, 34 bots)	**+322.94 USDT**	[Measured] ties to Weekly Review
Total return on 1,000 USDT	**+32.29%** (≈ +32.3% before 6.2 USDT withdrawal)	[Measured]
Closing balance (sheet)	1,316.74 USDT (after 6.2 withdrawn)	[Measured]
Average weekly return	5.79% (range 4.67% – 7.71%, 5 weeks)	[Measured]
Win rate (per bot)	**97.1%** (33 W / 1 L)	[Measured]
Profit Factor	**73.6** (327.39 gross wins / 4.45 gross loss)	[Measured]
Avg win / avg loss	+9.92 / −4.45 → 2.2 : 1 payoff	[Measured]
Expectancy per bot	+9.50 USDT (+3.38% mean, +1.81% median ROI)	[Measured]
Maximum drawdown (realized equity)	**0.00%**	[Measured] — **not** a true MDD (no MTM)
Sharpe (daily, ann. √365)	12.6	[Derived] — not decision-grade
Sortino / Calmar	Undefined (no negative day / zero MDD)	[Not computable]
Annualized ROI (simple)	≈ 368%	[Derived] — do not extrapolate
CAGR (compounded)	≈ 2,334%	[Derived] — arithmetic artefact of a 32-day window; **disregard**
Grid profit vs. asset appreciation	Cannot be split	[Not computable] — no price data
Fee impact	Not recorded	[Not computable]; modelled below

🏆 Golden Win

Weekly returns are unusually stable (4.7% – 7.7%) and the loss distribution is truncated: the single losing bot (ADA, −3.27%) cost 4.45 USDT, i.e. **1.4% of total profit**.

🚨 Critical Alerts

1. **Deployed notional peaks at 3,874 USDT (≈ 2.9× equity) on 2026-10-20.** Either the account uses leverage/margin, bots were run sequentially intra-day (dates have no time component), or investment figures are mis-logged. This must be resolved before any risk figure can be trusted. *(Upper bound — see Section II.)*
2. **Zero observable drawdown is a measurement artefact**, not evidence of low risk. Open positions and inventory marks are absent.
3. **Profit concentration:** NEAR/USDT = 42.6% of profit; the 7 bots ≥ 900 USDT = 48.1% of profit; the top 3 days = 41.5%.

Core Strategy Verdict

The program harvests mean reversion in mid-cap altcoin ranges with short holding periods (mean 1.2 days) and has produced a strong, steady realized yield with a very high hit rate. However, the sample covers one month, contains no recorded range break, and omits mark-to-market and fee data — so the observed Sharpe and zero drawdown reflect *harvest-then-exit discipline in a cooperative tape* rather than demonstrated tail resilience. **Viable and worth scaling cautiously; not yet underwritten for larger capital.**

Section II — Grid Microstructure & Operational Audit

Grid Parameter Review [Measured]

Parameter	Observed
Mode	100% Neutral (spot-style, two-sided)
Grid count	20 – 181 (median 40)
Range width (upper/lower − 1)	4.8% (WTI) → 188.7% (牛来); median 20.4%
Geometric step per grid (derived from range ÷ grids)	mean **0.50%**, min 0.18%, max 1.26%
Order distribution	Spacing type not logged; geometric step shown for comparability
Holding period	mean 1.18 d, median 1 d; **12 of 34 bots (35%) opened and closed same-day**
Bot size	17 bots < 150 USDT; 9 bots 150–500; 8 bots ≥ 500 (incl. 7 ≥ 900)

Grid spacing vs. ATR / volatility regime — [Not computable]. No price history is in the journal. Using step size as a proxy, bots with steps ≤ 0.25% (ADA 0.22%, TRX 0.18%, WTI 0.24%) sit at a fee-fragile spacing (see below). The ratchet in the NEAR lower bound across successive deployments (4.33 → 4.22 → 4.25 → 4.12 → 3.90 between 10/07 and 10/20) is consistent with a downward-drifting or expanding-volatility tape that you were re-centering manually.

Fee & Execution Drag Analysis

The journal records **Profit as `Final − Investment`**, i.e. already *net of whatever the venue charged*. Gross-vs-net therefore cannot be separated from the data. A model:

> **[Derived — assumption to verify]** Maker/taker on Pionex grid fills ≈ **0.05% per fill** → ≈ **0.10% round-trip** per completed grid cycle.

Geometric step	Round-trip fee as % of gross cell margin	Net cell margin
0.18% (TRX, 9/29)	**≈ 56%**	0.08%
0.22% (ADA)	**≈ 45%**	0.12%
0.50% (portfolio mean)	≈ 20%	0.40%
0.88% (NEAR typical)	≈ 11%	0.78%

Finding: The only losing bot (ADA, 181 grids on a 50% range) is also the tightest-spaced bot in the sample. That is the signature of **fee-dominated spacing** compounded by inventory carry in a trending leg. Rule of thumb: step ≥ 3× round-trip cost (≥ 0.30%). **Slippage drag:** [Not computable] — recommend logging fill-vs-limit and maker/taker split.

Inventory Management & Capital Efficiency

Metric	Value	Basis
Average deployed notional (calendar-day, inclusive of start/end dates)	≈ 921 USDT	[Derived], upper bound

Metric	Value	Basis
Average utilization vs. equity	≈ 80%	[Derived]
Zero-deployment days	3 of 32 (9/24, 9/25, 10/14)	[Measured]
Peak deployed notional	**3,874 USDT on 10/20 (≈ 2.9× equity)**; 2,577 on 10/10 (≈ 2.2×)	[Derived] — date resolution only
Return per unit capital, by pair (cumulative profit ÷ cumulative investment)	HYPE 6.5% · WLFI 10.0% · RENDER 4.0% · NEAR 3.6% · FRAX 3.9% · **INJ 0.84%** · **TRX 0.28%** · **COPPER 0.34%** · **WTI 0.45%**	[Measured]

Inventory skew: Neutral-mode grids start ≈ 50/50 base/quote and become **long-heavy as price falls** (each fill buys the dip). Asymmetric risk is therefore always to the downside: a neutral spot grid is short volatility on the lower tail and long-inventory at the boundary. No bot was reported out-of-range, but all positions were closed within ≤ 7 days — i.e. *the journal shows that risk was avoided by early exit, not that it was absent.*

Capital-allocation misalignment: TRX (1,222 USDT deployed → 3.46 profit) and INJ (1,998 → 16.84) absorbed ~27% of all capital-dollars for ~6% of profit, while HYPE/WLFI/RENDER produced 4–10% per deployment on much smaller tickets. Larger tickets are earning **lower** yield per dollar (≥ 500 USDT bucket: 2.1%; 150–500 bucket: 4.8%), which is an efficiency leak.

Section III — Risk & Stress-Testing Assessment

Boundary Risk Analysis

- **Recorded range breaks: none** [Measured as absent from journal]. Closed-bot data cannot show whether any bot sat below its lower bound for a period before being stopped.
- **[Derived] Value at lower bound.** Model: arithmetic grid, started at range midpoint, half in base asset, equal USDT per buy level, price pinned at the lower bound:

Bot	Range	Bot size	Modelled loss at lower bound
NEAR (10/10)	4.22 – 6.15	978.61	**−136.7 USDT (−14.0%)**
NEAR (10/20)	3.90 – 6.07	921.42	**−151.9 USDT (−16.5%)**
FRAX (10/10)	0.2716 – 0.345	999.80	−89.9 USDT (−9.0%)
TRX	0.322 – 0.349	1,073.29	−32.1 USDT (−3.0%)
INJ	8.212 – 8.876	998.87	−28.9 USDT (−2.9%)

Wide-range bots (NEAR) carry a *deep* inventory tail but earn the most per cycle; narrow-range bots (TRX, INJ) have shallow tails but also minimal yield. A single NEAR lower-bound event would erase **~3 weeks of the program's average profit**.

Volatility & Regime Sensitivity

Regime proxy	Observation
Wide range / 30–45 grids (NEAR, HYPE)	Highest yield per cycle (NEAR bots 1.6–6.1% ROI, 1–3 d holds) — **high-vol harvesting**
Tight range / low vol (TRX, INJ, WTI, COPPER, SOL)	ROI 0.1–1.1%; WTI 7-day hold earned 0.45% — **low-vol consolidation yields poorly relative to capital tied up**
Trend-expansion exposure	Only ADA (−3.27%) shows a trend-leg loss
Regime classification against ATR/ADX is [Not computable] without price data.	

Worst-Case Scenario Assessment [Derived, indicative]

Stack of bots concurrently open on 10/20 (NEAR 955, INJ 999, INJ 999, NEAR 921 — treat as an *upper bound* because of date resolution):

Shock	Approx. portfolio loss	% of ~1,300 equity
Alt basket −10% (avg delta ≈ 0.5–0.6 on 3,874 notional)	≈ −190 to −230 USDT	≈ −15% to −18%
NEAR to lower bound, INJ to lower bound	≈ −340 USDT	≈ **−26%**
Flash crash −30% through all lower bounds (delta ≈ 0.8): 1.0× notional → 2.9× notional	≈ −310 → −930 USDT	**−24% → −72%**

⚠ Risk Callout

If the 2.9× peak notional reflects real leverage/margin, a −35% simultaneous move on that stack is a **capital-impairment event**. If it reflects sequential same-day cycling, the tail is roughly one-third of that. *The journal cannot distinguish the two — resolve this first.*

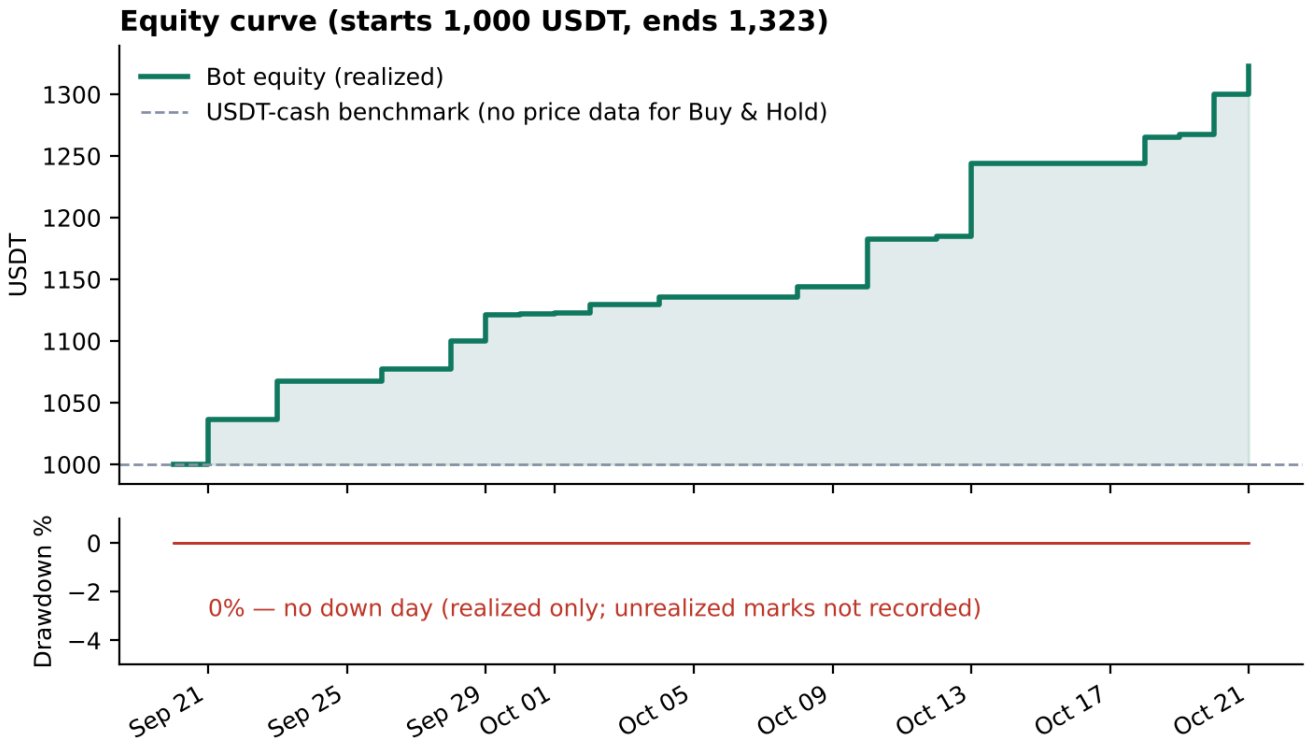
Divergence / Impermanent Loss vs. Buy-and-Hold

[Not computable] — requires entry/exit prices. Required fields: `entry price`, `exit price`, `open-bot unrealized value`. For reference, a neutral grid underperforms Buy-and-Hold whenever price trends beyond the range one-way; it outperforms in sideways/mean-reverting tape. Given 97% win rate over a 32-day window, the strategy mostly sampled the second regime.

Section IV — Data Visualizations

1. Equity Curve & Drawdown (realized, USDT; base 1,000, pre-withdrawal)

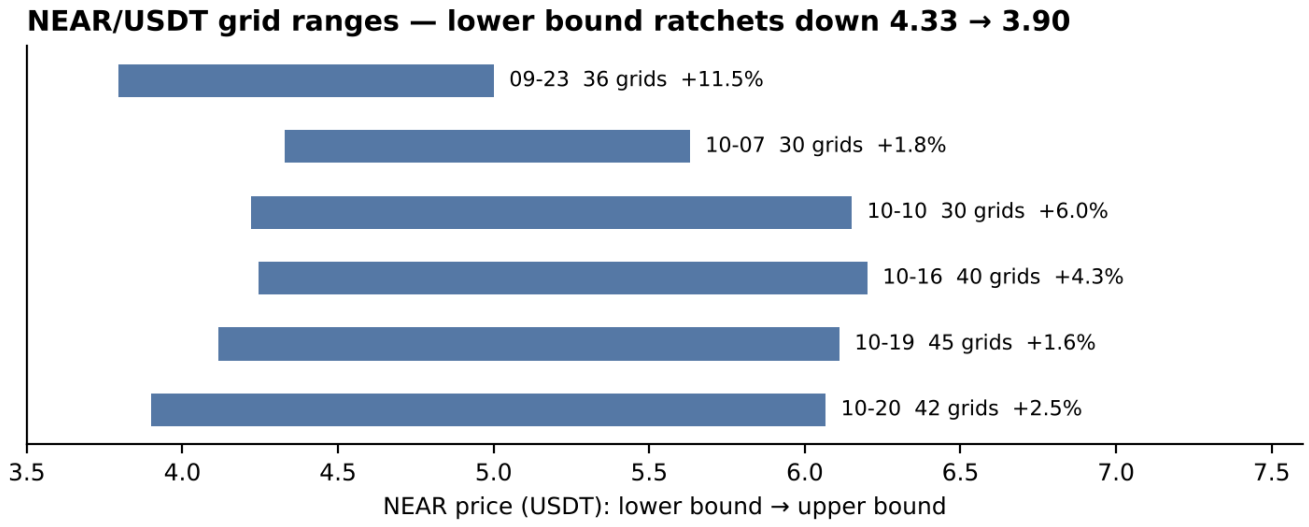
Buy-and-Hold benchmark: **not available** (no price data) — flat USDT-cash line (1,000) shown as the only benchmark.



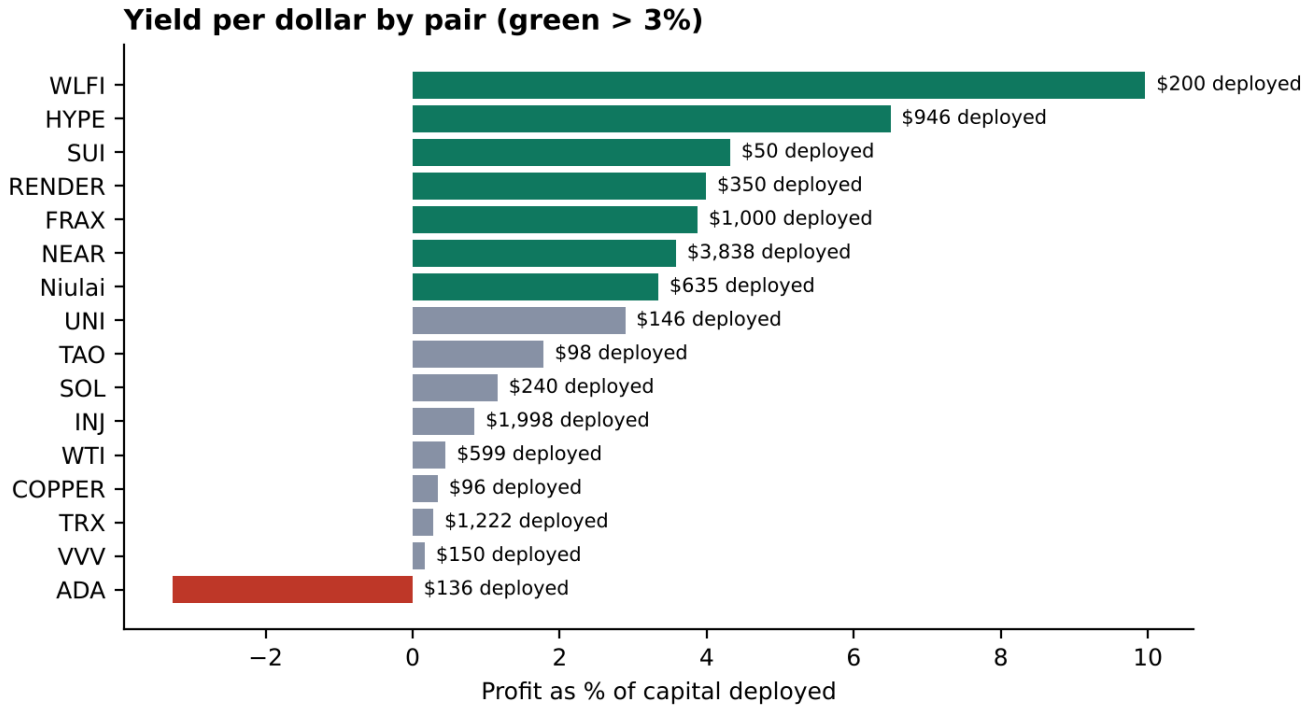
Realized equity vs a flat USDT-cash line. No Buy & Hold line is possible because the journal has no price data.

2. Range Map — NEAR/USDT deployments (the program's largest profit source)

Price axis in USDT, `[` lower bound, `]` upper bound, bars scaled 3.6 → 6.4.



Each bar is one NEAR bot: left edge = lower price, right edge = upper price.

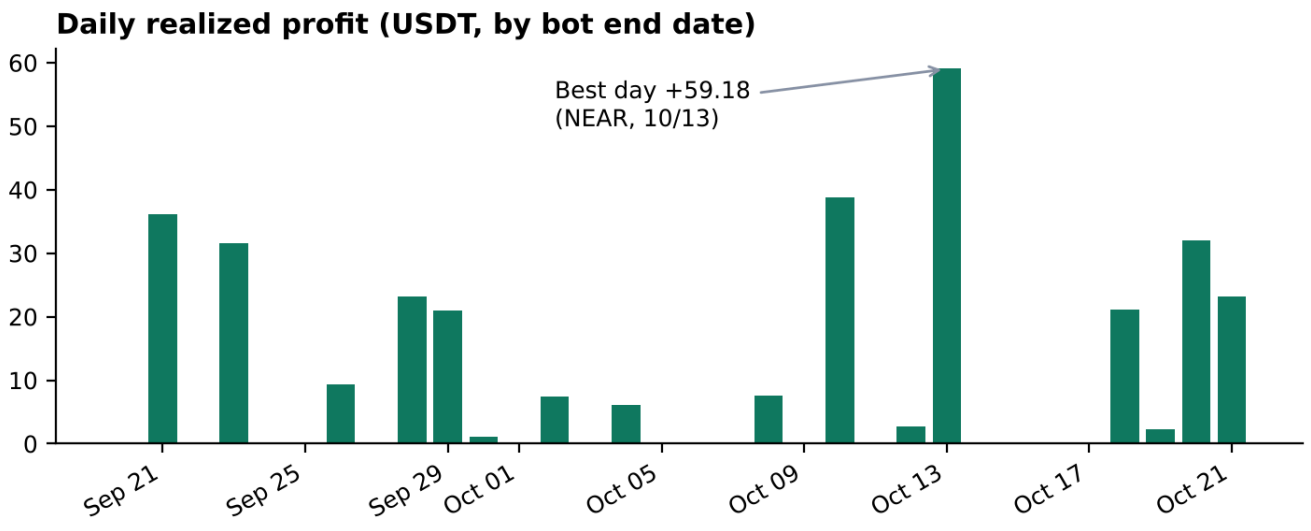


Which pairs earn the most per dollar tied up.

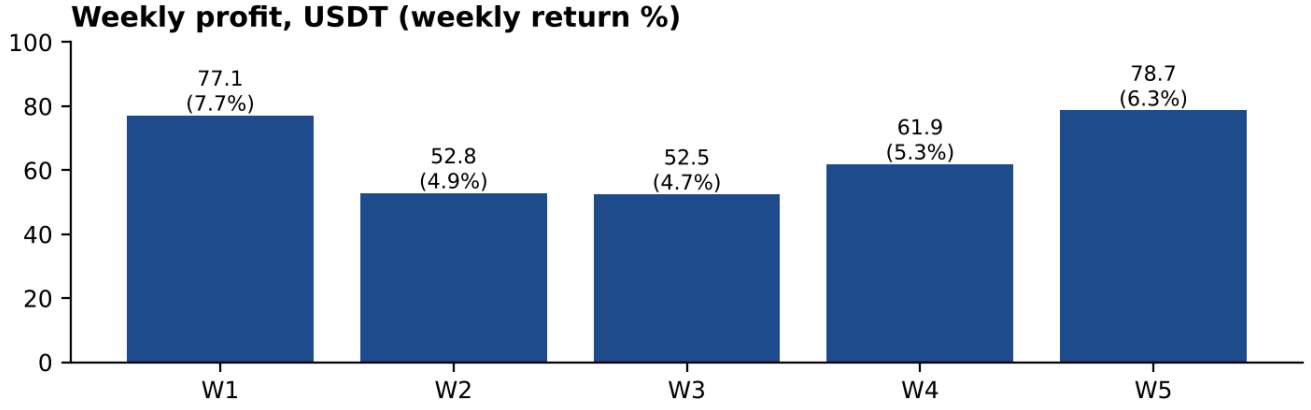
Trigger price and per-level fill density are not in the journal. Proxy for density — cycle yield per grid count — is tabulated below.

Pair	Bots	Capital deployed	Profit	Yield on deployed	Profit per grid-level (avg)
NEAR	6	3,838	+137.51	3.58%	0.62
HYPE	4	946	+61.51	6.50%	0.38
FRAX	1	1,000	+38.76	3.88%	0.86
牛来	3	635	+21.24	3.34%	0.05
WLFI	1	200	+19.93	9.97%	0.40
INJ	2	1,998	+16.84	0.84%	0.37
RENDER	4	350	+13.97	3.99%	0.07
TRX	3	1,222	+3.46	0.28%	0.04
ADA	1	136	**−4.45**	−3.27%	−0.02

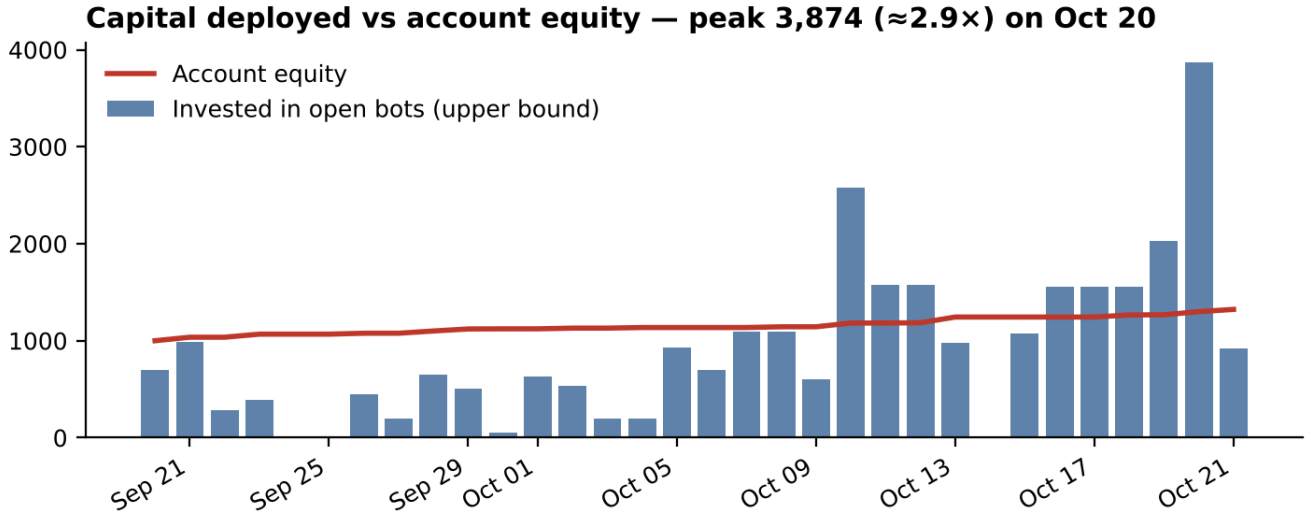
3. Daily PnL Distribution (realized, by bot end date; 1 block ≈ 1.5 USDT)



Most profit arrives in bursts when large bots close.



Weekly profit and weekly return.



Why the leverage / sequencing question matters.

Section V — Strategic Optimization & Recommendations

1. **Volatility-adjusted spacing (ATR integration).** Set `step = max(0.30%, 0.15 × ATR(14, 1h)/price)`; reject any configuration with step < 3× round-trip fees. This would have screened the only losing bot (ADA, 0.22%) and the TRX/COPPER/SOL bots whose yield on deployed capital is < 1.2%.
2. **Dynamic re-centering triggers.** The NEAR lower bound was already ratcheted down ~10% in 13 days; formalize it: re-center when price is within 10% of either bound *or* inventory skew > 70% to base; cap re-centers at 2/day to avoid chasing.
3. **Hard stop + trailing harvest.** (a) Stop-and-liquidate if price closes below lower bound −3% or bot mark-to-market ≤ −5% of ticket; (b) harvest when cumulative bot ROI ≥ 2× its rolling median (≈ 3.5%) or after 3 days, whichever first. Your own data shows holds ≤ 3 days produce nearly all profit.
4. **Perp delta-hedge for wide-range tickets.** For bots ≥ 500 USDT with ranges ≥ 30% (NEAR, FRAX), open a short perp equal to `0.5 × notional` at start and scale to track inventory delta; this converts the −14% to −16% lower-bound tail into funding/basis cost (est. 2–5 bps/day) while preserving grid harvest.
5. **Capital reallocation and portfolio caps.** Cap any single pair at 25% of equity and total deployed notional at ≤ 1.0× equity (or ≤ 1.5× if leverage is deliberate and documented). Rotate tickets from TRX/INJ (0.3–0.8% yield) to HYPE/RENDER/WLFI-class volatility; tickets of 150–500 USDT produced the best yield per dollar (4.8%).

Appendix A — Data-Integrity Findings (fix before the next audit)

#	Finding	Impact
1	**Row 1 is labelled `EXAMPLE ROW: overwrite`** (HYPE, +7.55) yet is included in Weekly Review totals	Total profit overstated by 2.3%; excluding it gives **315.39 USDT (+31.5%)**
2	Row 4 contains an orphan `FRAX/USDT` label with no data	Harmless to totals; cleanup
3	Bots 24–26 (BTR ×2, WTI) end 9/14–9/18, **before** the 9/20 tracking start (13.12 USDT)	Correctly excluded from Weekly Review (336.06 raw → 322.94 reported)
4	Bots 24–37 are logged out of chronological order	Reduces auditability
5	Workbook subtitle says **Bybit**; Setup says **Pionex**; Focus pair blank	Fee model depends on venue
6	**Peak concurrent investment (3,874) ≈ 2.9× equity**	Unresolved leverage / sequencing ambiguity
7	Dates have no time component; 12 bots show `Days Running = 0`	Intraday ROI/annualization impossible
8	Planned withdrawal 20%; actual **6.2 USDT = 8.0% of Week-1 profit**, none since	Compounding is stronger than plan
9	Status is Closed for all bots — no open-position marks	True MDD / unrealized loss invisible
10	Weekly Review shows 52 rows; only 5 weeks populated (forward weeks show 0% return)	Do not average them in

Appendix B — Fields to Add for a True 10/10 Audit

`Start time / End time (UTC)` · `Entry price` · `Exit price` · `Spacing type (Arithmetic/Geometric)` · `Fees paid` · `Maker/taker split` · `Grid profit vs. position PnL (Pionex shows both)` · `Max unrealized drawdown per bot` · `Leverage (if any)` · `Open-bot snapshots at each weekly review` · `Stop reason (target / manual / out-of-range)`.

Appendix C — Methodology

- Daily returns: realized profit attributed to bot **end date**, divided by running balance at the start of that day, 32 calendar days incl. zero-profit days.
- Sharpe = mean/σ × √365 (rf = 0); Sortino and Calmar undefined (no negative day, zero MDD).
- Geometric step = (Upper/Lower)^(1/Grids) − 1.
- Concurrency = sum of investments for all bots whose [start, end] date range contains the day (inclusive; upper bound).
- Lower-bound loss model: arithmetic grid starting at mid-range, 50% base / 50% quote, equal USDT per buy level, price pinned at the lower bound, fees ignored.
- Fee model (0.05% per fill) is an **assumption to verify** against the venue's current schedule.
- The 32-day sample (34 trades) is too small for inferential claims; treat ratios as descriptive only.